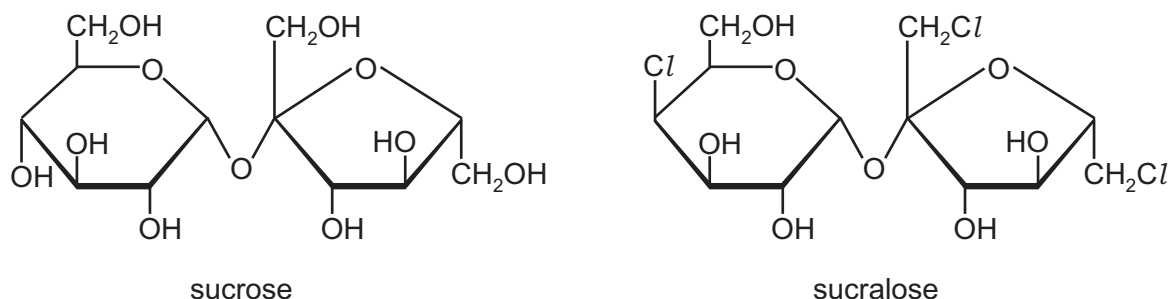


12 The diagram shows sucrose and sucralose.



The enzyme sucrase breaks down sucrose but cannot break down sucralose. Four students were asked to suggest why sucrase can break down sucrose but not sucralose.

Three of the students gave correct suggestions.

Which suggestion **cannot** be correct?

- A** The Cl atoms change the shape of the sucralose molecule so it is not the same shape as the active site of sucrase.
- B** The Cl atoms of the modified fructose cannot bind to the active site of sucrase.
- C** The Cl atoms cannot cause an induced fit, so sucralose does not enter the active site of sucrase.
- D** The Cl atoms cause fewer temporary hydrogen bonds between sucralose and the active site of sucrase.